



A Systematic Pre-Modeling Optimization Framework for Solar Power Forecasting Using Gradient Boosting Ensembles

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Motivation & Research Gap

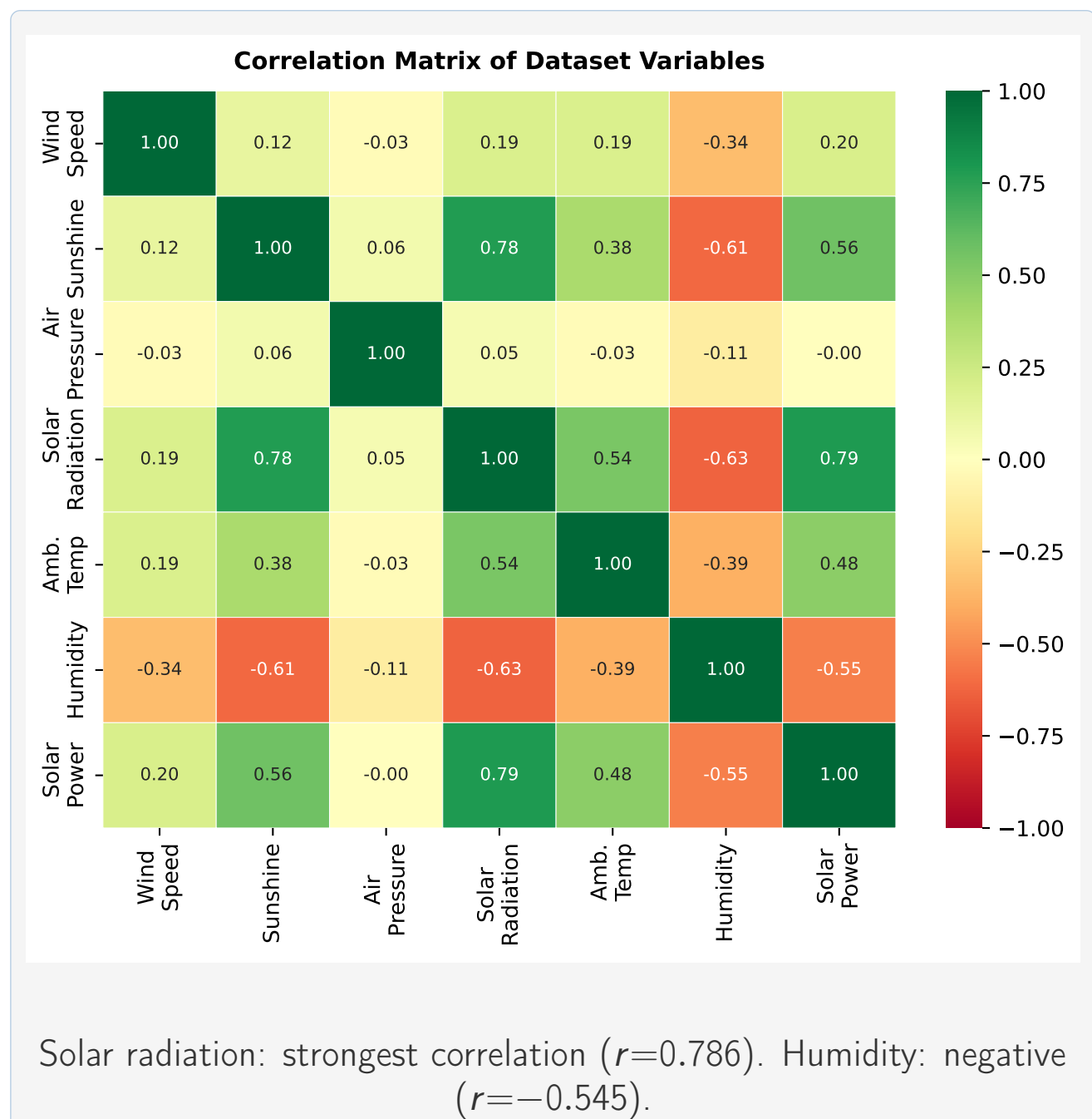
The literature gap: Prior work focuses on *algorithm comparison* (LR, ANN, LSTM, CNN-LSTM) while ignoring three systematic sources of suppressed accuracy:

- **Night-time zeros** — 62.6% of hourly records are trivially zero (no sunlight), distorting R^2 by design
- **Raw features only** — no cyclic time, physics interactions, or lag/memory features
- **Skewed target** — right-skew causes loss functions to over-penalize rare peak-hour errors

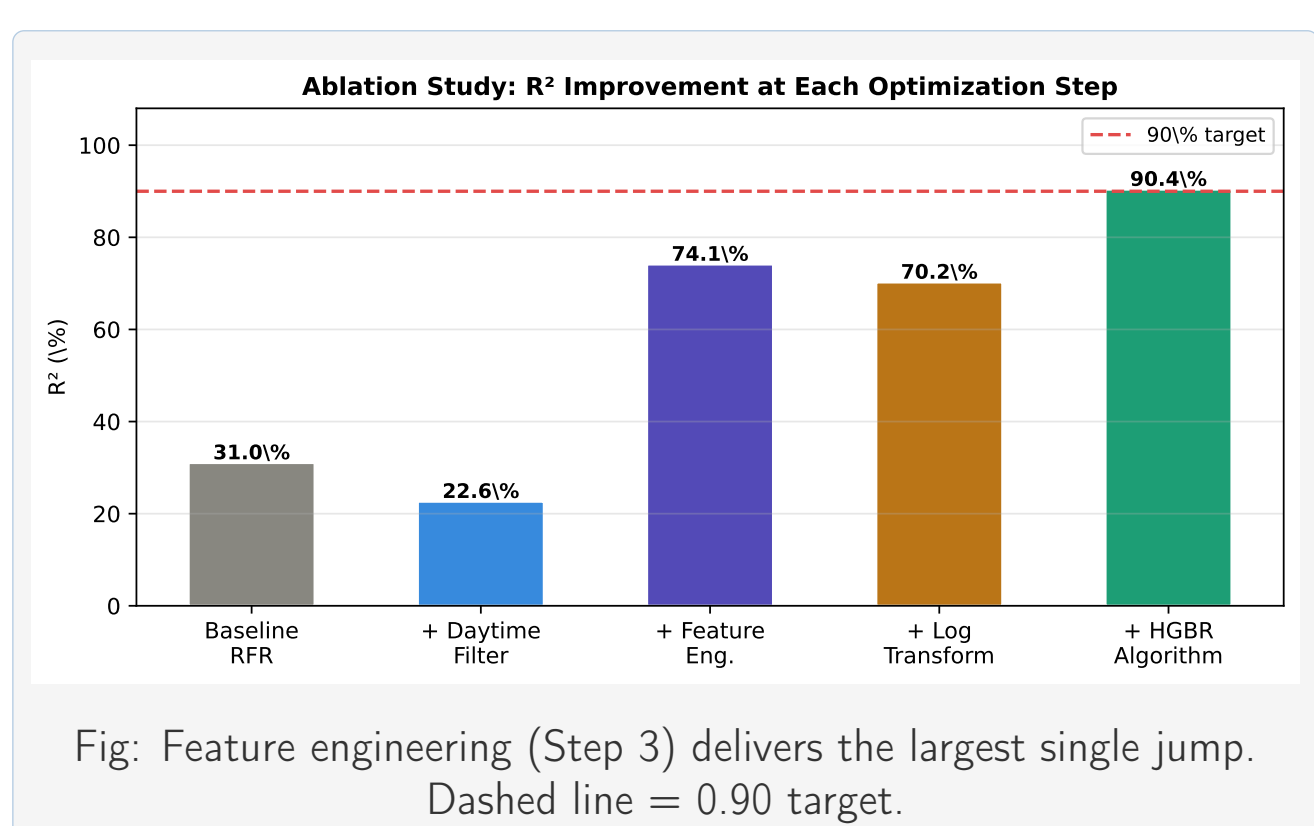
Research Questions

- RQ1: How much accuracy gain comes from pre-modeling decisions vs. any suitable algorithm selection?
- RQ2: Can structured feature engineering sustain $R^2 > 0.90$ across 1–10 day horizons (days ahead forecasting)?
- RQ3: Does HistGradientBoosting (HGBR) with log-transformed targets outperform published LSTM/RFR baselines?

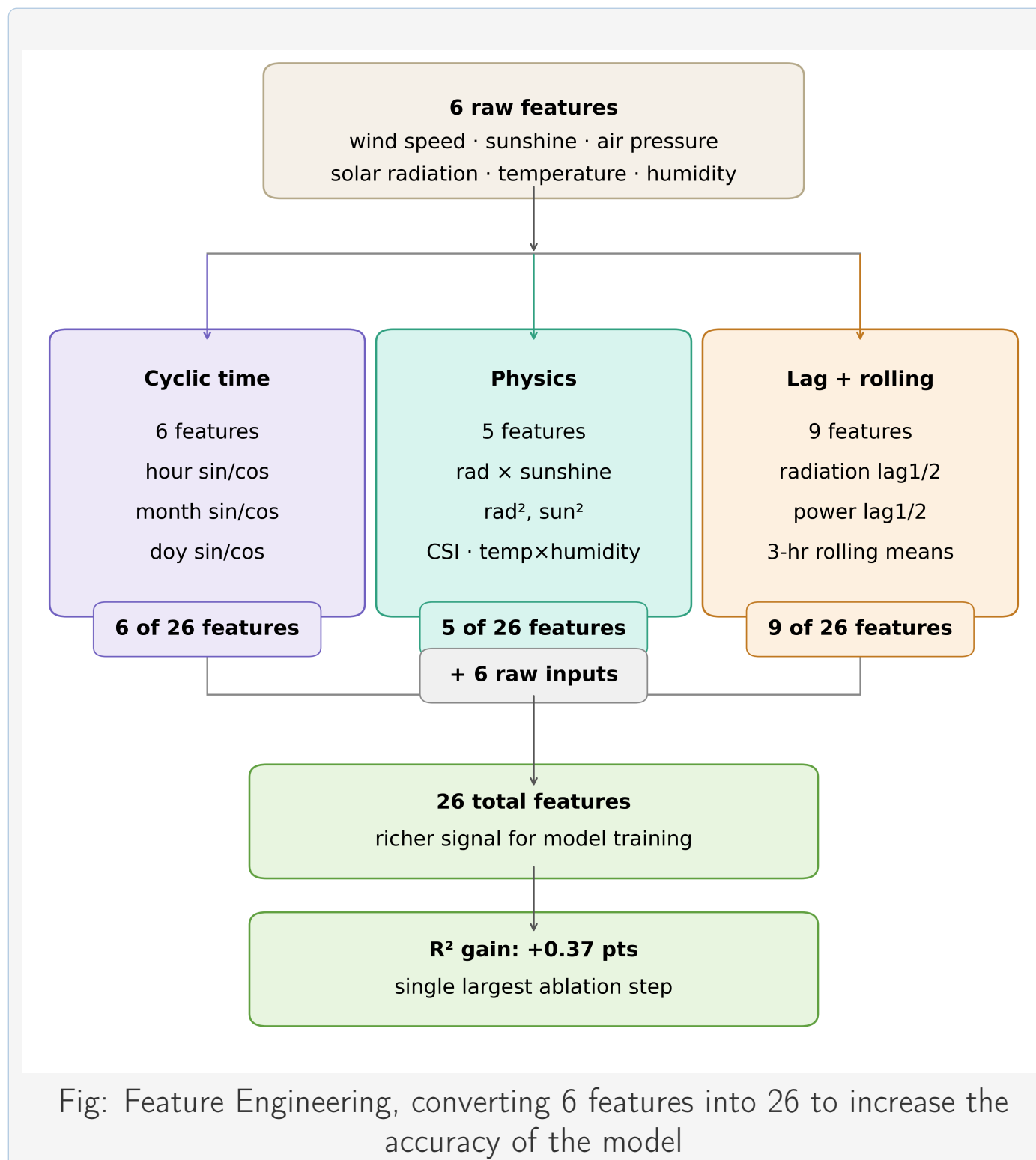
Correlation Matrix



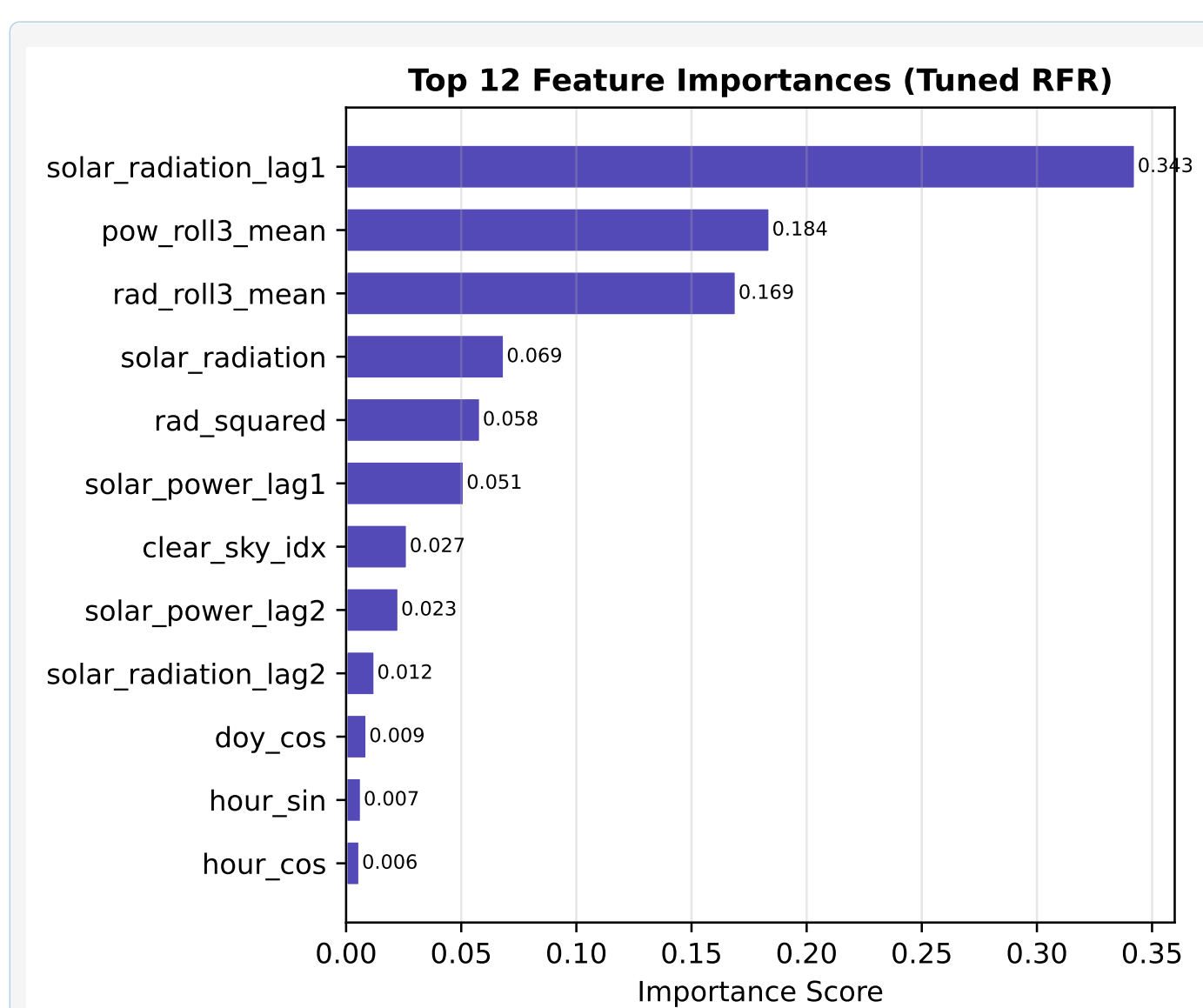
R^2 Improvement (Ablation)



Feature Engineering



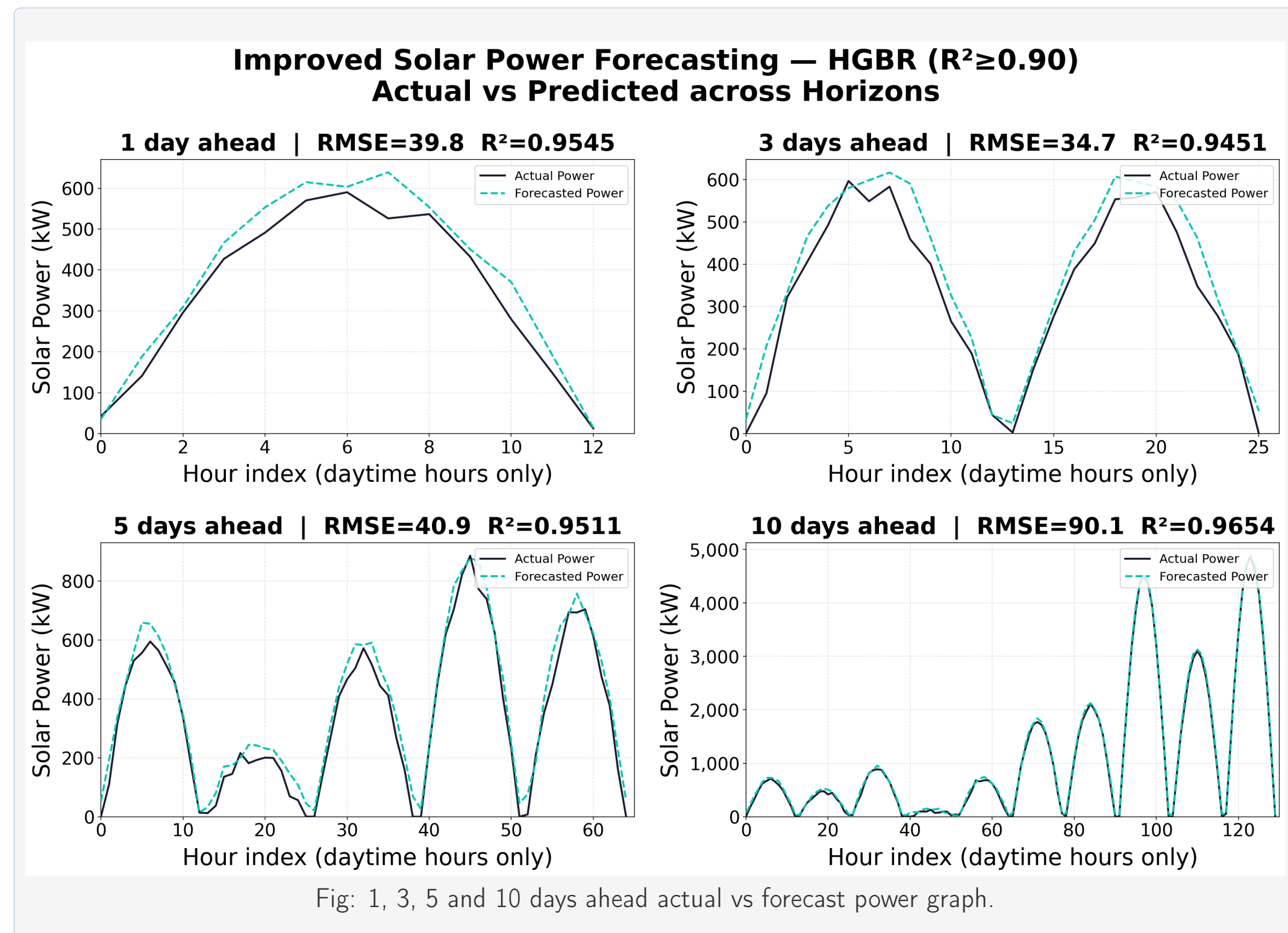
Feature Importance (Tuned RFR)



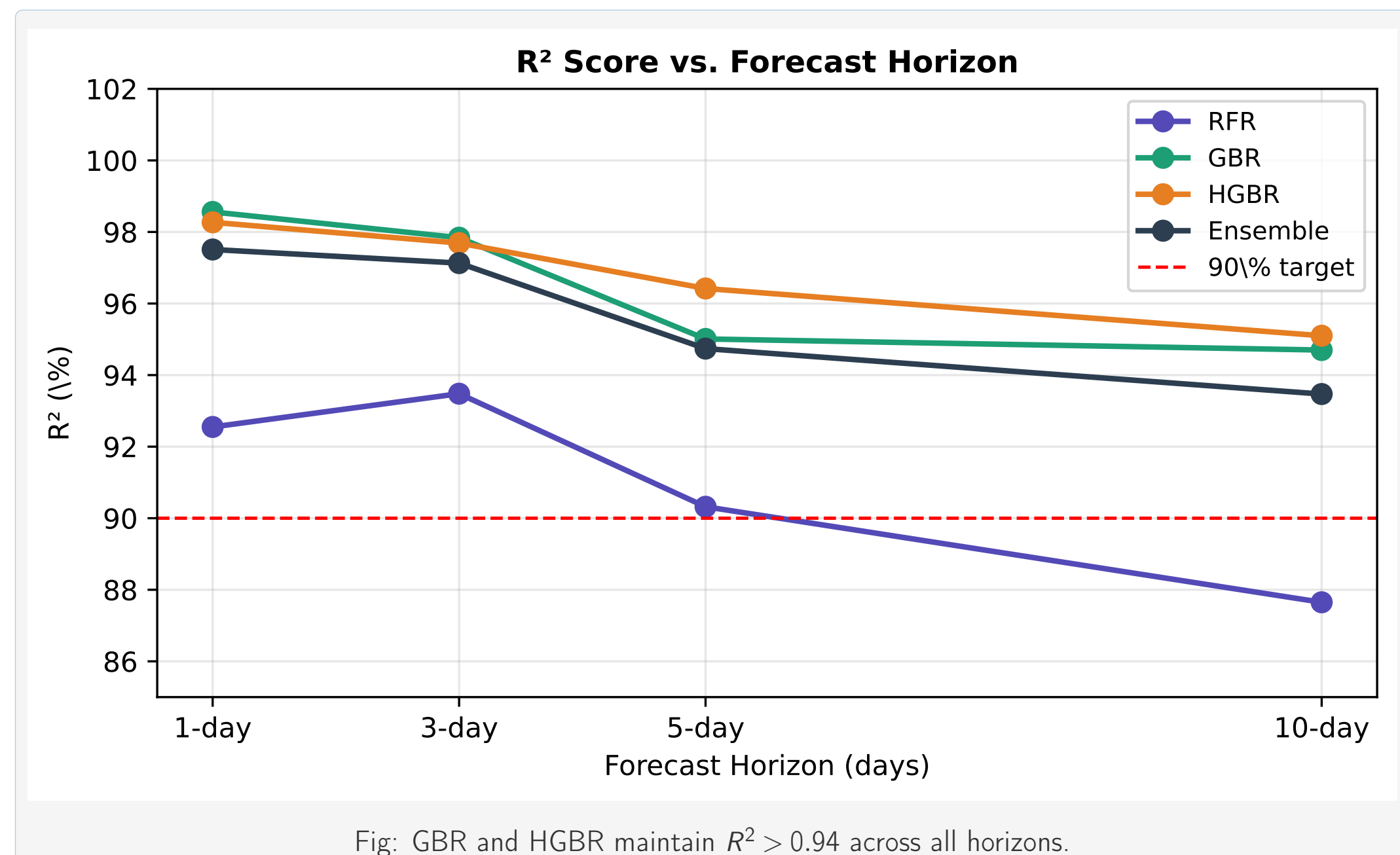
Conclusions

- ✓ Pre-modeling steps (filtering + feature engineering + log-transform) contribute **0.39 R^2 points** vs. 0.20 from algorithm upgrade — a **2:1 ratio**
- ✓ HGBR achieves $R^2 = 0.9037$ on full test set, outperforming published LSTM (≈ 0.91 – 0.92) without GPU infrastructure
- ✓ All models sustain $R^2 > 0.90$ at 1-, 3-, 5-day horizons; HGBR reaches **0.951 at 10 days**
- **Future work:** SHAP explainability, transfer learning across geographies, quantile boosting for probabilistic forecasts

Forecasting HGBR for 1 to 10 Days Ahead



R^2 vs. Forecast Horizon



References

1. Balal et al., *IEEE GreenTech* 2025
2. Kim et al., *Electr. Power Syst. Res.*, 2023
3. Lim et al., *Energies*, vol. 15, 2022
4. Mystakidis et al., *Computing*, 2023
5. Yagli et al., *Sol. Energy*, vol. 208, 2020
6. Inman et al., *Prog. Energy Combust. Sci.*, 2013

